PROBLEMS

7-1. The register of Fig. 7-1 transfers the input information into the flip-flops when the *CP* input goes through a positive-edge transition. Modify the circuit so that the input information is transferred into the register when a clock pulse goes through a negative-edge transition, provided a load input control is equal to binary 1.

7-2. The register of Fig. 7-3 loads the inputs during a negative transition of a clock pulse. What internal changes are necessary for the inputs to be loaded during the positive edge of a pulse?

7-3. Verify the circuit of Fig. 7-5 using maps to simplify the next-state equations.

7-4. Design a sequential circuit whose state diagram is given in Fig. 6-27 using a 3-bit register and a 16×4 ROM.

7-5. The content of a 4-bit shift register is initially 1101. The register is shifted six times to the right, with the serial input being 101101. What is the content of the register after each shift?

7-6. What is the difference between serial and parallel transfer? What type of register is used in each case?

7-7. The 4-bit bidirectional shift register of Fig. 7-9 is enclosed within one IC package.

(a) Draw a block diagram of the IC showing all inputs and outputs.

(b) Draw a block diagram using three ICs to produce a 12-bit bidirectional shift register.

7-8. The serial adder of Fig. 7-10 uses two 4-bit shift registers. Register A holds the binary number 0101 and register B holds 0111. The carry flip-flop Q is initially cleared. List the binary values in register A and flip-flop Q after each shift.

7-9. What changes are needed in the circuit of Fig. 7-11 to convert it to a circuit that subtracts the content of B from the content of A?

7-10. Design a serial counter; in other words, determine the circuit that must be included externally with a shift register in order to obtain a counter that operates in a serial fashion.

7-11. A flip-flop has a 20-ns delay from the time its *CP* input goes from 1 to 0 to the tune the output is complemented. What is the maximum delay in a 10-bit binary ripple counter that uses these flip-flops? What is the maximum frequency the counter can operate at reliably?

7-12. How many flip-flops must be complemented in a 10-bit binary ripple counter to reach the next count after 0111111111?

7-13. Draw the diagram of a 4-bit binary ripple down-counter using flip-flops that trigger on the (a) positive-edge transition and (b) negative-edge transition. positive edge : Q——> clock negative edge: Q'——>clock

7-14. Draw a timing diagram similar to that in Fig. 7-15 for the binary ripple counter of Fig. 7-12.

7-15. Determine the next state for each of the six unused states in the BCD ripple counter of Fig. 7-14. Is the counter self-starting?

7-16. The ripple counter shown in Fig. P7-18 uses flip-flops that trigger on the negative-edge transition of the *CP* input. Determine the count sequence of the counter. Is the counter self-starting?

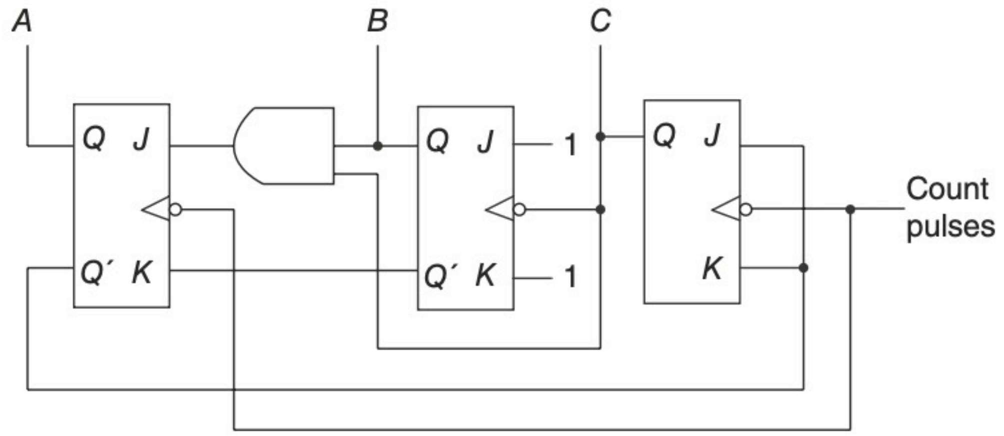

Figure P7-18 Ripple counter

7-17. What happens to the counter of Fig. 7-18 if both the up and down inputs are equal to 1 at the same time? Modify the circuit so that it will count up if this condition occurs.

7-18. Verify the flip-flop input functions of the synchronous BCD counter specified in Table 7-5. Draw the logic diagram of the BCD counter and include a count-enable control input.

7-19. Show the external connections of four IC binary counters with parallel load (Fig. 7-19) to produce a 16-bit binary counter. Use a block diagram for each IC.

7-20. Construct a BCD counter using the MSI circuit of Fig. 7-19.

7-21. Construct a mod-12 counter using the MSI circuit specified in Fig. 7-19. Give four alternatives.

7-22. Using two MSI circuits as specified in Fig. 7-19, construct a binary counter that counts from 0 to binary 64.

7-23. Using the *stop* variable from Fig. 7-21 as a start signal, construct a second word-time control that stays on for a period of 16 clock pulses.

7-24. Show that an n-bit binary counter connected to an n-to-2^n line decoder if equivalent to a ring counter with 2^n flip-flops. Draw the block diagrams of both circuits for $n = 3$. How many timing signals are generated?

7-25. Include an enable input to the decoder of Fig. 7-22(b) and connect it to the clock pulses. Draw the timing signals that are now generated at the outputs of the decoder.

7-26. Complete the design of the Johnson counter of Fig. 7-23, showing the outputs of the eight timing signals.

7-27. (a) List the eight unused states in the switch-tail ring counter of Fig. 7-23. Determine the next state for each unused state and show that, if the circuit finds itself in an invalid state, it does not return to a valid state, (b) Modify the circuit as recommended in the text and show that (1) the circuit produces the same sequence of states as listed in Fig. 7-23(b), and (2) the circuit reaches a valid state from any one of the unused states.

7-28. Construct a Johnson counter with ten timing signals.

7-29. (a) The memory unit of Fig. 7-24 has a capacity of 8192 words of 32 bits per word. How many flip-flops are needed for the memory address register and memory buffer register?

(b) How many words will the memory unit contain if the address register has 15 bits?

7-30. When the number of words to be selected in a memory is too large, it is convenient to use a binary storage cell with two select inputs; one X (horizontal) and one Y (vertical) select input. Both X and Y must be enabled to select the cell.

(a) Draw a binary cell similar to that in Fig. 7-29 with X and Y select inputs.

(b) Show how two 4×16 decoders can be used to select a word in a 256-word memory.

7-31. (a) Draw a block diagram of the 4×3 memory of Fig. 7-30, showing all inputs and outputs.

(b) Construct an 8×3 memory using two such units. Use a block diagram construction.

7-32. It is required to construct a memory with 256 words, 16 bits per word, organized as in Fig. 7-33. Cores are available in a matrix of 16 rows and 16 columns.

(a) How many matrices are needed?

(b) How many flip-flops are in the address and buffer registers?

(c) How many cores receive current during a read cycle?

(d) How many cores receive at least half-current during a write cycle?

Differentiate between Ripple counter and synchronous counter.

Write short notes on

(a) Registers

(b) Johnson counter

(c) Ring counter

(d) Random access memory

(e) Parallel in Serial Out (PISO) Shift registrar

7-35. Design a synchronous counter up-down with JK flip-flop which counts the odd number from 1 to 15 in binary.

7-36. Design a sequential circuit whose table is given bellow

Present Sate Input			Next State		Out put
A_1	A_2	x	A_1	A_2	y
0	0	0	0	1	0
0	0	1	1	1	0
0	1	0	1	0	1
0	1	1	1	1	1
1	0	0	1	1	1
1	0	1	0	1	0
1	1	0	1	0	0
1	1	1	0	0	0

7-37. Repeat the problem 4 with ROM and a register.

7-38. Design a MOD 10 synchronous binary up counter using T flip-flop and other necessary logic gates. Draw the timing diagram.

7-39. Design a 4-bit up/ down asynchronous counter using JK flip-flops and other necessary logic gates. Use one directional control input. If P = 0, the counter will count up and for P = 1, the counter will count down.